

Administrative Information

1. Title

Modulation of antihypertensive drug effectiveness by dietary consumption: a scoping review

2. Registration

The protocol for this review will be registered on Zenodo.

3. Authors

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4. Amendments

Any important protocol amendments will be documented with a rationale and date of amendment.

5. Support

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Introduction

6. Rationale

Effective medical treatment of hypertension is of utmost relevance as approximately 1 in 3 adults between 30-79 years old suffer from it, a mere 54% has been formally diagnosed and only around 21% of them has adequate blood pressure control. Hypertension is defined as having a systolic blood pressure (SBP) of ≥ 140 mmHg or diastolic blood pressure (DBP) of ≥ 90 mmHg or taking medication for hypertension (World Health Organization, 2023). The American Heart Association has adopted a more conservative definition of ≥ 130 mmHg SBP and ≥ 80 mmHg DBP to account for complications that can arise at lower numbers and to allow early intervention (Whelton et al., 2018). Importantly, large-scale meta-analyses of randomised clinical trials (RCTs) with normotensive and cardiovascular disease patients, respectively, demonstrates that a reduction of 5 mmHg of SBP decreases the risk of major cardiovascular events by approximately 10% (Pinho-Gomes et al., 2021; Rahimi et al., 2021).

There are various types of potent antihypertensive drugs available. First line drugs include angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor blockers (ARBs), calcium channel blockers (CCBs) and diuretics. Generally, these primary drug types provide similar cardiovascular protection, and (contra)indications per drug type for comorbidities are taken into account at prescription. ACEIs, ARBs and direct renin inhibitors interfere with the renin-angiotensin system (RAS), which is physiologically responsible for regulating BP, fluid and electrolyte balance and vascular resistance. CCBs inhibit calcium from entering the muscle cells of the heart and arteries, thereby inhibiting contraction, lowering heart rate and BP. Diuretics increase the urinary outflow of water and salt, thereby preventing increase in blood volume and BP. Among other drugs used against hypertension are beta blockers, which directly affect the heart rate, the heart's workload and output, decreasing BP. Alpha blockers function by reducing arterial resistance and relaxing the muscle tone of

arterial walls. These drugs are prescribed in stepwise commencement as monotherapy and combination therapy to control BP (Laurent, 2017; Williams et al., 2018).

Despite the potency of the various types of antihypertensive drugs, suboptimal control of BP remains a widespread issue (Abegaz et al., 2017; World Health Organization, 2023). Causes of suboptimal control of BP include inadequate prescription of suitable drugs and incomplete adherence to drugs by patients (Burnier & Egan, 2019; van der Linden et al., 2021). Adherence problems often stem from side effects of the medication, exacerbated by comorbidities, although both are taken into account at prescription as per guidelines. Here, we noticed there is barely attention for the role of dietary consumptions on the effectiveness of and compliance with antihypertensive drugs (Williams et al., 2018). Interference with the functionality of drugs may be in the form of acute interaction due to simultaneous intake or more chronic ineffectiveness of the drug due to habitual dietary intake. Additionally, there might be subgroups that have a higher risk of suffering from food-drug interactions, such as people with comorbidities and/or poor diets (Otles & Senturk, 2014). Mechanistically, food-drug interactions occur due to changes in intestinal transport, systemic distribution, metabolism, and excretion. A well-known food-drug interaction occurs mainly by the enzymatic system known as cytochrome P450 (CYP), and the isoform CYP3A4 specifically is responsible for the inactivation and removal of 50-60% of drugs (Boullata & Hudson, 2012).

A comprehensive evaluation and overview of dietary items, including supplements, and interference with the effectiveness of antihypertensive drugs is lacking. Consequently, healthcare givers are not guided to take patients' dietary habits into account when prescribing and advising medication (Williams et al., 2018). Herein, we recognised a potential contributing factor to suboptimal BP control in patients who take medication. This scoping review aims to identify specific dietary items and diets that may modulate the effect of antihypertensive drugs. By identifying dietary items interesting for further study, this review will pave the way for the incorporation of the role of diet in the prescription of antihypertensive therapy.

7. Objectives

Research question: Which dietary consumptions modulate antihypertensive drug effectiveness?

The question includes food, drink and supplemental compounds and dietary habits that may influence the effectiveness of antihypertensives. The review will register findings both on the effects of simultaneous intake of specific food and beverages with antihypertensives as well as the effects of dietary habits.

PICO

Participants: The review will include research wherein the participants were adults and took antihypertensive drugs for either hypertension or as normotensives, and not solely for other morbidities.

Intervention: Participants take antihypertensive drugs, and the dietary items under study are registered.

Comparators: Compare with different diet habits or control group.

Outcomes: Altered drug efficiency, side effects, change in BP response

Methods

8. Eligibility Criteria

Study characteristics

- Study population is human adults treated with antihypertensives
- Time frame: prospective, antihypertensive drug use registered before outcome
- Primary research
- Included study designs: cohort study, RCTs including crossover studies, case cohort, nested case control
- Excluded study designs: case control, cross-sectional, animal studies
- Excluded study settings: antihypertensive drug use for other conditions, non-CVD conditions with specialised medical and dietary care
- Baseline specific antihypertensive drug type usage available
- Specific dietary consumption available

Report characteristics

- Published anytime
- English and Dutch language, abstract available in English
- Full publications

9. Information sources

PubMed/Medline databases will be used to look for studies to include in the review.

10. Search strategy

A systematic search will be conducted to answer the research question using the search string below. Manual search may also be conducted if required.

((food*[Title/Abstract] OR diet*[Title/Abstract] OR nutri*[Title/Abstract] OR beverage*[Title/Abstract] OR drink*[Title/Abstract] OR consum*[Title/Abstract] OR intake[Title/Abstract] OR ingest* [Title/Abstract] OR digest*[Title/Abstract] OR snack*[Title/Abstract] OR phosph*[Title/Abstract] OR coffee*[Title/Abstract] OR caffeine [Title/Abstract] OR tea[Title/Abstract] OR supplement*[Title/Abstract] OR dairy*[Title/Abstract] OR milk [Title/Abstract] OR yoghurt [Title/Abstract] OR yogurt[Title/Abstract] OR vitamin [Title/Abstract] OR cholecalciferol*[Title/Abstract] OR mineral[Title/Abstract] OR minerals[Title/Abstract] OR potassium*[Title/Abstract] OR sodium* [Title/Abstract] OR salt*[Title/Abstract] OR oil [Title/Abstract] OR “omega 3” [Title/Abstract] OR botanical*[Title/Abstract] OR grapefruit*[Title/Abstract] OR eat[Title/Abstract] OR eating[Title/Abstract] OR licorice[Title/Abstract] OR liquorice[Title/Abstract] OR St. John’s Wort[Title/Abstract] OR magnesium[Title/Abstract] OR alcohol*[Title/Abstract] OR ethanol [Title/Abstract] OR iron[Title/Abstract] OR ferrous fumarate[Title/Abstract] OR star coin[Title/Abstract] OR Panax quinquefolius[Title/Abstract] OR Allium sativum[Title/Abstract] OR garlic[Title/Abstract] OR Salvia miltiorrhiza[Title/Abstract] OR Silybum marianum[Title/Abstract] OR Curcuma longa[Title/Abstract] OR turmeric[Title/Abstract] OR Echinacea[Title/Abstract] OR Ginkgo biloba[Title/Abstract] OR Hypericum perforatum[Title/Abstract] OR Camellia sinensis[Title/Abstract] OR valerian* [Title/Abstract] OR cranberr* [Title/Abstract] OR kava[Title/Abstract] OR danshen[Title/Abstract] OR sage [Title/Abstract] OR ginseng[Title/Abstract] OR “milk thistle” [Title/Abstract] OR broccoli[Title/Abstract] OR sprout*[Title/Abstract] OR cabbage*[Title/Abstract] OR fiber*[Title/Abstract] OR fibre*[Title/Abstract] OR “black cohosh”[Title/Abstract] OR actaea

racemosa [Title/Abstract] OR saw palmetto [Title/Abstract] OR goldenseal[Title/Abstract] OR leafy[Title/Abstract] OR cheese*[Title/Abstract] OR chocolate*[Title/Abstract] OR yohimbine[Title/Abstract] OR hawthorne[Title/Abstract] OR roselle[Title/Abstract] OR guggul[Title/Abstract] OR piperine[Title/Abstract] OR capsicum[Title/Abstract] OR cannabidiol*[Title/Abstract] OR "folic acid"[Title/Abstract] OR gelatin[Title/Abstract] OR hemp*[Title/Abstract] OR hibiscus[Title/Abstract] OR probiotic*[Title/Abstract] OR zinc[Title/Abstract] OR cocoa[Title/Abstract] OR polyphenol[Title/Abstract] OR flavonoid*[Title/Abstract] OR isoflavone*[Title/Abstract] OR lignan[Title/Abstract] OR soy[Title/Abstract] OR phytoestrogen*[Title/Abstract] OR nutraceutical* [Title/Abstract] OR herb* [Title/Abstract] OR berr* [Title/Abstract] OR wine* [Title/Abstract] OR beer* [Title/Abstract])

AND ("Antihypertensive Agents" [Mesh] OR "antihypertensive agent*" [Title/Abstract] OR anti-hypertensive* [Title/Abstract] OR antihypertensive* [Title/Abstract] OR "hypertension drug" [Title/Abstract] OR "hypertension drugs" [Title/Abstract] OR "blood pressure drug" [Title/Abstract] OR "blood pressure drugs" [Title/Abstract] OR "blood pressure medication" [Title/Abstract] "beta blocker" [Title/Abstract] OR "beta blockers" [Title/Abstract] OR "alpha blocker" [Title/Abstract] OR "alpha blockers" [Title/Abstract] OR diuretic* [Title/Abstract] OR ACE[Title/Abstract] OR angiotensi* [Title/Abstract] OR ARB[Title/Abstract] OR calcium channel[Title/Abstract] OR vasodilator* [Title/Abstract] OR "mineralocorticoid receptor antagonist*" [Title/Abstract])

11. Study records

Data management

After the search, retrieved results will be screened with the assistance of ASReview. ASReview is an open-source title and abstract screening tool which uses AI to learn which types of reports should be included and excluded and then accelerates the screening process by sorting by relevance. This allows reviewers to be more efficient as a significant percentage of the retrieved reports will not need manual screening (van de Schoot et al., 2021). After screening the remaining studies will be assessed full text for eligibility based on the inclusion criteria. The reason of exclusion will be recorded for the excluded studies. Data from the final included studies will be extracted and recorded in a pre-made Excel data-extraction sheet.

Selection process

The screening, assessment of eligibility and inclusion will be conducted by the first and second author.

Data collection process

Data from the studies included will be extracted and recorded in a pre-made Excel data-extraction sheet.

12. Data items

Author names, year of publication, source of funding, journal of publication, countries where the study was conducted, study design, study duration, study period, study population: number, sex, patient type/general, mean age, intervention type, intervention duration and period, follow-up time, exposure, main outcome variables, baseline BP, drug type, results, BP post intervention, BP change, C_{max}, AUC, T_{max}, P-values, hazard ratios, comorbidities, adjusted for, study conclusions, bias in reporting and conclusions.

13. Outcomes and prioritization

Change or difference in SBP and DBP, change or difference in maximum serum concentration (C_{max}) of drug, change or difference in drug bioavailability measured in area under the curve (AUC).

14. Risk of bias in individual studies: inapplicable for scoping reviews

15. Data synthesis: inapplicable for scoping reviews

16. Meta-biases: inapplicable for scoping reviews

17. Confidence in cumulative evidence: unnecessary for scoping reviews

18. References

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